Seonghun **Bae**

ROBOTICS & FUTURE MOBILITY LEARNER

Nam-gu, Ulsan, Republic of Korea

(+82) 10-9598-0665 | baeseonghun@naver.com | https://blog.naver.com/seonghun-s | seonghun00

"Fail forward, grow upward."

Summary

This is Seonghun Bae, who wants to become robotics engineer seeking to specialize in autonomous navigation and mobile robot (AMR) control systems. My main interests are in ROS2, SLAM, Navigation, simulation and mobile robot control systems. I always try to be a humble learner, listening to feedback from senior engineers and growing together with my colleagues. To improve my skills every day, I carefully record and share my project journeys on GitHub, always ready to take on new challenges and learn from experience.

Research Interests

SLAM / Navigation Interest in 2D/3D SLAM, Nav2, and autonomous path planning.
Simulation Building robot environments and testing programs in Gazebo, and MuJoCo.

Education

Yeungnam University *Gyeongsan, S.Korea*
B.S. IN ROBOTICS ENGINEERING *Mar. 2022 - Feb. 2026*

- Total GPA : 3.79 / 4.5
- Club : ROS Study Group - Academic Member & Project Lead

Projects

Outdoor Mapping Data Optimization using a Quadruped Robot *Yeungnam University*
PARTICIPANT *Nov. 2024 - Dec. 2025*

- Using Deep Robotics Lite3 quadruped robot to make outdoor environment maps - [Github]
- Testing mapping algorithms and making data better in outdoor environments.
- Fixing sensor errors and getting clean 2D/3D map data from real outdoor roads.
- Learning how to handle 4-legged robot hardware and balance it on uneven ground.

Making a Hand Gesture Recognition Model *Yeungnam University*
PARTICIPANT *Mar. 2025 - Jun. 2025*

- Reviewed ICCVW papers to study computer vision for object detection algorithms. - [Github]
- Developed an on-device hand tracking pipeline using MediaPipe to extract 3D landmarks and embedding vectors.
- Built a custom dataset (50+ images per class) to train a classifier and improve recognition accuracy.
- Implemented a real-time system to detect and classify custom gestures immediately from live video streams.

Development of a Mobile Robot with Person Following System *ROS Study Group*
PARTICIPANT *Apr. 2024 - Sep. 2024*

- Designed and assembled the mobile robot's hardware platform and chassis using aluminum profiles.
- Implemented an object detection system using an RGB-D camera to recognize and bounding-box a target person.
- Integrated Arduino board with motor drivers to control the hoverboard wheel motors for actual robot driving.
- Tested the overall system interoperability by verified real-time camera tracking and hardware movement controls.

Skills

Programming C++, C, Python, Matlab
Robotics & DevOps ROS, Linux, Git, Docker
Languages Korean, English

Certificates

Jun. 2026	**Engineer Information Processing (Written)**, HRDK (Human Resources Dev. Service)
Mar. 2026	**ABEEK Graduate (Washington Accord)**, ABEEK Korea
Mar. 2026	**Robotics Professional (Accredited)**, Korea Robotics Society
Nov. 2025	**TOEIC Speaking AL (170)**, YBM
Apr. 2022	**Computer Literacy (Grade 2)**, KCCI (Chamber of Commerce)
Dec. 2018	**Type 1 Normal Driver's License**, Korean National Police Agency

Extracurricular Activity

GitHub Study Log Repository
Github

OPEN SOURCE CONTRIBUTOR — *Apr. 2026 - Present*

- Operating a public GitHub repository to document daily learning logs - [GitHub]
- Focusing on studying advanced robotics technologies and recording the codes continuously.
- Making study logs open to everyone to share robotics knowledge with global peers.
- Reviewing and refactoring my own source codes to build better software habits.
- Trying to learn core robot systems every day and growing as a passionate engineer.

Technical Blog
Naver Blog

TECHNICAL BLOGGER — *April. 2026 - Present*

- Writing and managing a technical blog to share new engineering knowledge - [Blog]
- Summarizing lectures, reviewing technical papers, and posting helpful development tips.
- Sharing what I have studied to help other developers and remember information longer.
- Contributing to the tech community and making a better society by sharing free knowledge.
- Improving technical writing skills and learning how to explain complex ideas easily.

ROS Study Group
Yeungnam University

PROJECT LEAD & ACADEMIC ORGANIZER — *Apr. 2024 - Dec. 2025*

- Leading and organizing weekly ROS2 study meetings for robotics students.
- Managing team projects and helping members solve hardware and software problems.
- Sharing study materials and robot source codes on GitHub to help the team grow.
- Studying core robotics technology like SLAM, Nav2, and simulation programs together.
- Learning how to collaborate with other software and hardware engineers effectively.